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POWERED

FOSSIL

WRIST PDA: First Wearable PALM OS Device

Fossil Overview

- **Fossil is new entrant to technology, but long-standing leader in fashion accessories:**
  - Watches
  - Leather goods (handbags, wallets, belts)
  - Sunglasses
- **Fossil not only produces its own branded product but other brands:**
  - Emporio Armani
  - DKNY
  - Diesel
  - Also… Columbia, Billabong, Paul Frank, & Phillipe Starck
- **Fossil is publicly traded (NASDAQ:FOSL) with $663M in sales in '02**

Wrist TECHNOLOGY

- **Wristwatch is the only widely acceptable wearable device in world today**
- **Fossil believes mainstream personal communication & computing is destined for wrist wearable design**
- **Wearable devices must not only be functional but fashionable**



Wrist PDA™ System Specifications

- Palm OS Version 4.1.2
- 160 x 160 pixel LCD touch screen
  - 1, 2, and 4 bit per pixel grayscale
- Stylus Integrated into the watchband
- 33Mhz DragonballVZ processor
- IRDA port
- Rechargeable battery
- On screen inking - JOT Recognition
- USB Interface for HotSync
- Third Party Application Support



AWARDED BEST OF COMDEX 2002

The power of a Palm OS handheld on your Wrist!

AUTHENTIC FOSSIL®

Abacus WRIST PDAs



$179



$199

Fossil WRIST PDAs



$275



$295



$295

Wrist PDA™ Inputs/Outputs



Wrist PDA™ Palm OS System Changes

- **Three-way Rocker and Back buttons for one-handed user operation.**
- **Palm OS system forms, alerts, menus, and dialogs use larger Fossil fonts and objects.**
- **Larger Fossil fonts and larger form objects available to developers to create Wrist PDA optimized applications.**
- **New Auto-Off modes including a “Watch” mode with user customizable faces; third-party *watch applications*.**

Three-Way Rocker and Back Button

In order to be more compatible with current Palm OS devices and with future versions of Palm OS, the Wrist PDA uses the same virtual characters that Palm OS 5.2 has defined for "Jog Dial" like event definitions.

The Wrist PDA system has been modified to utilize the Rocker and Back buttons in menu and pop-up trigger lists. The Rocker and Back button control navigation, selection and exiting for these objects.

Controlling Back Button Behavior

- **The default behavior of the Back button is to return to the launcher. For a Wrist PDA application that wishes to use the Back button to "back-out" of a dialog or for some other purpose, the default behavior must be overridden.**
- **An API is provided to turn off the default behavior of the Back button so that the system will not process the request.**
- **The developer should remember to allow the application to exit! (Remember, there is no launcher button)**

How the larger Fossil Fonts and Objects Work

For each standard Palm OS font, There is a larger Fossil font equivalent:

stdFont → FossilStdFont

boldFont → FossilBoldFont

…

largeBoldFont → FossilLargeBoldFont

If the form is a *Fossil Form*, this 1-to-1 mapping allows the system to *automatically* change the font IDs of all the objects in the form to the corresponding larger Fossil font. The sizes of these objects are then recalculated to accommodate the larger font.

Question: What is a “Fossil Form”?

Answer: A Fossil Form is a Palm OS form that has been flagged by the developer to be drawn using the corresponding larger Fossil fonts.

To flag a form, the developer places a special gadget in the form. This gadget must have the following characteristics:

- An object ID terminating with "70” (decimal).
- A rectangle with the following coordinates:
  X = 1, Y = 2, Width = 3, Height = 4
- The usable bit must be turned OFF

Effects of the Fossil Form Gadget

When the Fossil gadget is detected by the system, several changes are made to the form:

- **The font IDs of all the objects in the form are mapped to the corresponding Fossil font.**
- **The height of List Boxes, Menus, and other objects are recalculated for the larger font size.**
- **The form title height is increased and the title is displayed in the corresponding Fossil font.**

Runtime Control of Object Size

However, most resource compilers do not allow you to specify the width and height of objects. To display larger Fossil fonts, most Form objects need to have their width and/or height recalculated.

An API call is provided to allow the developer to recalculate the correct size of objects at run time.

- Void FossilFrmEnlargeObject(FormType* fP, UInt16 objectIdx);
- The argument ***kFossilFrmEnlargeObjectAll*** may be used as a special value of the object index to enlarge all objects in the form.

Menus with Fossil Fonts in Absence of a Form

Menus do not have to be associated to a form. For this reason, a Fossil Menu API will allow the developer to manipulate whether a menu is displayed using Fossil fonts in the absence of a Form.

- void FossilMenuSetActiveMenuRscID(UInt16 ID,
  Boolean largerMenu);
- Boolean FossilMenuLargeFontGet(MenuBarType* mP);
- Boolean FossilMenuLargeFontSet(MenuBarType* mP,
  Boolean largerFont);

Will non-Wrist PDA Aware Applications Run?

If an application is not Wrist PDA aware (no Fossil API calls, no Fossil Gadget, uses standard fonts), then the application should operate normally.

Every time an application switch takes place, the Wrist PDA system is reset to basic Palm OS mode. This helps to ensure that the system state is always compatible with running applications.

LCD Refresh Rate Control

To help conserve battery, the Wrist PDA runs in a 5 frames per second mode (counted in system ticks). The developer can use the following API to change this value to suit the needs of the application. This value will return to the default 5fps when an application switch takes place.

- void FossilDisplayRefreshRateSet(UInt16 newRefreshRate);
- UInt16 FossilDisplayRefreshRateGet(void);

The Wrist PDA™ "Watch" Mode

- **Instead of a blank screen, the Wrist PDA can launch a "Watch" application when Auto-Off timeout is reached.**
- **The current time is displayed until the user enters into PDA mode by pressing Rocker Enter.**
- **The watch application is refreshed once every minute to keep time accurately while saving battery life.**
- **This is the mode the watch will be most of the time. *There is enormous opportunity for this real estate!***



3rd Party Watch Applications

- **3rd party developers will be able to create custom watch applications for the Wrist PDA platform.**
- **The end user can customize which watch application will be launched through the Wrist PDA preference panel.**
- **Full details on API Specification for Watch Applications will be available soon.**



AUTHENTIC FOSSIL

Additional Information

Fossil will be at the lab and in a booth during the conference. CD-ROMs with the Alpha release of the SDK will be available in these locations while supplies last.

Visit the developers sections of the product websites to download new SDK:

- www.Fossil.com (***see Fossil Tech section***)
- www.AbacusWatches.com

Enter your e-mail address at one of these websites to receive notifications about the SDK.